

MD. ZAHIDUL HAQUE (ALVI)

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Current Ph.D. Student · Software Engineering Researcher · AI/ML Engineer

EDUCATION

- **William & Mary** Williamsburg, VA, USA
Ph.D. Student in Computer Science (Current) Aug 2023 - Present
 - **Status:** Active Ph.D. Student in the AURA Lab. Expected Graduation: May 2028.
 - **Research Focus:** Intersection of AI and Software Engineering, specializing in **Parameter-Efficient Fine-Tuning (PEFT)** for **Large Code Models**.
 - **Relevant Coursework:** Software Maintenance & Evolution, Deep Learning, AI for Software Engineering, Autonomous Driving & Connected Mobility, Deep Transfer Learning, Data Mining, Human-Centered Computing.
- **Sylhet Engineering College (SEC)** Sylhet, Bangladesh
Bachelor of Science in Computer Science & Engineering; GPA: 3.46/4.00 Mar 2016 - Mar 2021

PUBLICATIONS

- **Parameter-Efficient Multi-Task Fine-Tuning in Code-Related Tasks**
Md. Zahidul Haque, Saima Afrin, Antonio Mastropaolo
Submitted to ACM Transactions on Software Engineering and Methodology (TOSEM), Jan 2026.
📄 [Preprint] 🐙 [Code]
 - Investigated **Multi-task QLoRA** fine-tuning across code generation, translation, and summarization tasks. First study to analyze both functional correctness and non-functional code quality in parameter-efficient multi-task settings.
- **A Systematic Literature Review of Parameter-Efficient Fine-Tuning for Large Code Models**
Saima Afrin, Md. Zahidul Haque*, Antonio Mastropaolo*
ACM Transactions on Software Engineering and Methodology (TOSEM), Oct 2025.
📄 [Paper] 🐙 [Code] (*Equal contribution)
 - Analyzed the efficacy of **PEFT** techniques (including **LoRA and Adapters**) on large-scale code models (CodeLlama, StarCoder) to improve adaptability in software engineering tasks.

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, CUDA, Bash
- **AI/ML Frameworks:** PyTorch, TensorFlow, **LangChain**, **PEFT (LoRA/QLoRA)**, Hugging Face Transformers
- **Data & Tools:** Pandas, NumPy, Scikit-learn, Git, Docker, Linux/Unix Environments, Jupyter

PROFESSIONAL EXPERIENCE

- **Graaho** Reston, VA (Remote)
Software Engineer (Machine Learning) Mar 2022 - Sep 2022
 - **Credit Risk Modeling:** Engineered a financial machine learning pipeline for credit risk assessment; performed extensive data cleaning, implemented baseline modeling using Scikit-learn, and established testing protocols with Pytest.
- **Mayalogy** Dhaka, Bangladesh
Research Engineer (Machine Learning) Feb 2021 - Feb 2022
 - **Biomedical NER Pipeline:** Developed “MayaNERC”, a pipeline to identify 17 biomedical entities from patient labels. Validated performance across 111 diseases categorized into five major classes (Men’s health, Pregnancy, etc.).
 - **NLP Tools Development:** Built “Auto Tag”, “Group Referrer”, and a Spam Filter for Romanized Bangla and Urdu text, achieving Macro F1 scores of 85%, 89%, and 88% respectively.
 - **Team Leadership:** Led an AI annotation team of 20 engineers and 15 psychologists to curate high-quality Deep Learning datasets; collaborated with a medical board of 150+ doctors to enhance AI-assisted healthcare solutions.

RESEARCH PROJECTS

- **B-NER: A Novel Bangla Named Entity Recognition Dataset** IEEE Access, 2023
First Author · [\[Paper\]](#) [\[Code\]](#) Apr 2023
 - **Dataset:** Created the largest Bangla NER dataset with 22,144 annotated sentences and 8 entity types from newspapers and Wikipedia.
 - **Evaluation:** Achieved 86% Macro-F1 using fine-tuned IndicBERT; conducted cross-dataset validation proving superior generalization.

TEACHING EXPERIENCE

- **William & Mary** Williamsburg, VA
Graduate Teaching Assistant Fall 2023 - Present
 - **Courses:** Generative Artificial Intelligence for Software Development (CSCI 455), Prompt Engineering (CSCI 100), Principles of Programming Languages (CSCI 312), Intro to Programming (CSCI 141).
 - **Responsibilities:** Conducted lab instructions for 30+ students, graded assignments, and provided office hour support for Python and C++ curriculum.

HONORS COMPETITIONS

- **Robi Datathon 2.0** Remote
Pre-Assessment on Tabular Data Analysis Jun 2022
- **DL Sprint - BUET CSE Fest 2022** Dhaka, Bangladesh
Bengali Automatic Speech Recognition Competition Jul 2022
- **Cisco Certified Network Associate (CCNA)**
Active Certification 2020 - 2023